# Assignment – Full Stack Developer

## 1. SQL Search Bug

* **Where:** TaskRepository.java (the same mistake is in search-tasks.sql) and the oracle file
* **How I found it:** I Searched "api" and old archived tasks like "legacy API cleanup" were showing. They should be hidden.
* **Root Cause:** The query used AND and OR without brackets. In SQL, AND runs before OR. So the query worked like "(not archieved AND title matches) OR (description matches AND status matches)". An archieved task could still come in through the description match, and the status filter was skipped for title matches.
* **FIX:** I put brackets around the OR part: (title LIKE ... OR description LIKE ...). Now every condition must be together.

## 2. Slow Search

* **Where:** TaskController.java
* **How I found it:** Reading the code, I saw Thread.sleep inside the search method. Short or empty searches felt slow.
* **Root Cause:** The code paused for (10 minus query length) * 100 milliseconds. So an empty search waited 1 second for no reason.

* Fix : I deleted the sleep. It added delay and had no real purpose.

3. Loading stuck and old results

* Where : useTasks.js (frontend hook)

* How I found it : Reading the code, I saw setLoading (false) only ran on success. Also, if you type fast, two requests run and the slower one can arrive last.

* Root Cause : On an error, loading never became false, so the page showed "Loading" forever. The old error was also never cleared. An old response could replace newer results (a race condition).

* Fix : I added a cancelled flag in the effect cleanup so old responses are ignored. I clear the error at the start, and I use .finally() to always set loading to false.

4. Too many requests and wrong page

* Where : App.jsx

* How I found it : I saw that a request is sent for every letter typed. If I was on page 2 and searched, the result could be empty.

- Root Cause : The search text went straight to the API call, and the page number stayed the same when the filter changed
- FIX : I added a 300ms debounce so it waits until I stop typing. I also reset the page to 1 when the search or status changes.